AfAM Project Report

Compliant Nut Cracker

A Three-Level Adjustable Kitchen Tool Adapted from a Compliant Compression Mechanism

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AT A GLANCE

Mechanism	Compliant Compression Mechanism
Product	Adjustable Nut Cracker
Material	PLA — FDM printed
Parts	Two
Nut Sizes	Small · Medium · Large (3 slot levels)
Course	MN 203 DfAM

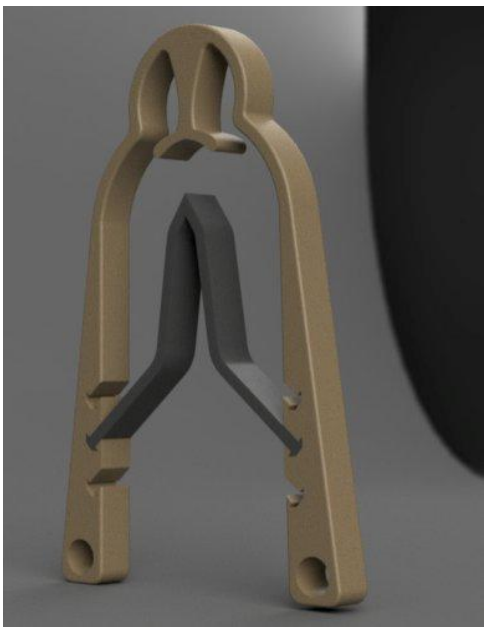

Fig. 1 — CAD render of the compliant nut cracker

01 · MECHANISM ORIGIN & FUNCTION

The source mechanism for this project is the Compliant Compression Mechanism. In its original topology, two slender flexible arms are connected at the top through a curved compliant arch. When the two handle arms at the bottom are pressed toward each other, the gap between two flat opposing faces at the top reduces, generating a controlled compressive force at the output.

The mechanism operates entirely through elastic deformation — there are no pins, hinges, or sliding contacts. Energy is stored in the bent arms and the deformed arch, and the structure returns to its open position the moment the applied force is released. This passive restoring behaviour, combined with its two-parts assembly, makes it exceptionally well-suited for a hand-held kitchen tool that must open and close repeatedly without mechanical wear.

02 · IDENTIFIED NEED & PRODUCT CONCEPT

Cracking hard-shelled nuts such as walnuts and almonds at home typically requires either a purpose-built rigid metal cracker or striking with a hard surface — both of which risk crushing the kernel or sending shell fragments flying. A compliant nut cracker addresses these issues by providing a controlled, progressive squeeze force that fractures the shell without shattering the edible interior.

The identified need is: a lightweight, single-piece, hand-held nut cracker that accommodates common nut sizes found in an Indian household — small (almond, ~10 mm), medium (dry coconut piece / hazelnut, ~18 mm), and large (walnut, ~28 mm) — without the user needing to swap parts or adjust screws. Three recessed slot levels machined into the jaw faces serve as passive size-stop positions, automatically limiting jaw closure depth and preventing over-cracking.

03 · AFAM DESIGN DECISIONS

3.1 Functional Adaptation

- The compression mechanism's output gap was oriented vertically so the nut sits naturally by gravity between the two jaw faces, centred without the user's active positioning effort.
- The arch at the top was thickened and profiled into a smooth dome to act as the primary structural load path during cracking, while remaining compliant enough for elastic deflection.
- The inner jaw faces were given three progressively deeper recessed notch levels (small / medium / large) that act as passive size stops — the nut self-seats in the appropriate level depending on its diameter.

3.2 Ergonomic & Aesthetic Choices

- The handle arms were flared outward at the grip ends and given a gentle taper, allowing three or four fingers to wrap naturally around each arm for a secure palm squeeze.
- Circular holes were included at the base of each handle — both to reduce material and weight, and to allow hanging storage on a kitchen hook.
- The overall silhouette was kept symmetrical and organic, resembling a wishbone or praying-hands form, making the product visually distinctive and immediately communicating its squeezing function to a new user.

3.3 DfAM-Specific Decisions

- The entire cracker is a two-part PLA print —, no fasteners, but with dovetail joints. The compliant flexure is integral to the part, which is the key advantage of additive manufacturing for compliant mechanisms.
- The part was oriented vertically on the print bed so that the thin compliant arch (highest-stress region) has layer lines running along its length rather than across, maximising inter-layer bond strength in the critical flexure zone.
- Wall thickness of the arms was set to 4 mm with 25% gyroid infill, balancing stiffness (enough to transmit cracking force) against flexibility (elastic return without plastic deformation).

04 · FABRICATION & TESTING

Fig. 2 — Prototype at hand scale; open rest position

Fig. 3 — Almond seated at small slot level, pre-crack

Fig. 4 — Walnut loaded at large slot level, being squeezed

Fig. 5 — Successfully cracked almond kernel extracted

4.1 Test Results

- Almond (small, ~10 mm): Seated cleanly in the small slot. A moderate two-finger squeeze fractured the shell along its seam. Kernel extracted intact in two halves. No shell scatter.
- Walnut (large, ~28 mm): Seated in the large slot. A firm full-palm squeeze produced an audible crack. Shell split into three to four fragments; kernel remained mostly whole and was easily separated.
- Passive return: In both cases, the cracker sprang back to its fully open position immediately upon releasing grip force, confirming that the compliant arch did not undergo plastic deformation during testing.
- Slot guidance: The recessed slot levels successfully prevented over-travel — the nut bottomed out in the slot before the arms could close further, protecting the kernel from being pulverised.

05 · CONCLUSION

The Compliant Nut Cracker demonstrates how a foundational compliant mechanism topology — the symmetric compression mechanism — can be adapted into a useful, elegant, and manufacturable kitchen product through thoughtful AfAM design. By leveraging additive manufacturing's ability to produce compliant structures, the design eliminates all moving parts, fasteners while delivering a tool that is ergonomic, size-adaptive, and self-restoring. The three-level jaw slot system is a simple but effective DfAM feature that requires no additional components and emerges naturally from the geometry of the printed part. Successful cracking of both almonds and walnuts in physical testing validates the design concept and the suitability of PLA as the sole material for this application.